

Relative Covariance

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This article describes how the phenotypic covariance between two individuals i and j can be modeled. Rather than assuming a fixed degree of relatedness from the outset, we treat the covariance as a quantity we *approximate given whatever information we happen to have about the pair*, and build up a hierarchy of increasingly strong assumptions — a “tower” — that lets us express it at four levels of resolution: from the exact, effect-weighted genetic cross-moment (Level 0), down to a function of the pair’s pedigree degree (Level 3). Each assumption is introduced only at the point the derivation requires it, is boxed and given a stable identifier (e.g. [\[A3\]](#)) so it can be referred to later, and is annotated with the common terminology that usually implies it. The intent is that more complex trait architectures (shared environment, assortative mating, population structure, dominance) can restart this same derivation from Level 0 and re-introduce a subset of these — or newly identified — assumptions.

1 Additive genetics and environment, without assortative mating

We begin with the simplest architecture: a trait determined by an additive genetic component and an independent environmental component, in a randomly mating population.

1.1 The phenotype model

For an individual i we write the (standardized) phenotype as

$$y_i = g_i + e_i, \tag{1}$$

where g_i is the genetic value and e_i the environmental deviation. We take y to be standardized across the population, $E[y] = 0$ and $\text{Var}[y] = 1$, and both components to be mean-centered, $E[g] = E[e] = 0$, with variances $\text{Var}[g] = V_A$ and $\text{Var}[e] = V_E$.

[A1] Additive genetic architecture.

The genetic value is a linear combination of the individual's standardized genotypes at the causal variants,

$$g_i = \sum_{k=1}^M \beta_k x_{ik},$$

where β_k is the per-standardized-allele causal effect of variant k , x_{ik} is the standardized genotype ($E[x_{ik}] = 0$, $\text{Var}[x_{ik}] = 1$), and M is the number of causal variants.

Commonly bundled into: the additive model; no dominance and no epistasis.

[A2] Independent, unshared environment.

The environmental component is independent of the genetic component and independent across individuals:

$$g \perp e \quad \text{and} \quad e_i \perp e_j \quad (i \neq j).$$

Commonly bundled into: no shared/common environment (C); no gene–environment correlation or interaction.

Because $g \perp e$, the phenotypic variance decomposes as $\text{Var}[y] = V_A + V_E$; with the standardization $\text{Var}[y] = 1$ this gives $V_A + V_E = 1$, so V_A is exactly the narrow-sense heritability, $V_A = h^2$. Every covariance we derive below is therefore also, numerically, a phenotypic *correlation* between the two individuals.

1.2 What we are computing, and conditional on what

For a single fixed pair, y_i and y_j are two numbers, so “their covariance” only has meaning as an average over an ensemble of pairs. The ensemble we use is the set of all pairs that share a given piece of *relatedness information* $\mathcal{I}_{ij}$ about the pair, and the object of interest is the conditional cross-moment

$$C_{ij}(\mathcal{I}) := E[y_i y_j \mid \mathcal{I}_{ij}]. \tag{2}$$

Reading (2) carefully: the expectation is taken across all pairs of individuals that match the conditioning information $\mathcal{I}_{ij}$ — *not* across all pairs (that average is ≈ 0 , since two randomly drawn individuals are uncorrelated).

Two remarks fix the interpretation:

- When $\mathcal{I}_{ij}$ is a statement about *descent* (which alleles are identical-by-descent, the genome-wide IBD proportion, or the pedigree degree), it carries no information about *which* alleles an individual actually carries, so it does not shift either marginal mean: $E[y_i \mid \mathcal{I}] = 0$. Then $C_{ij}(\mathcal{I}) = \text{Cov}[y_i, y_j \mid \mathcal{I}]$ exactly, and we may freely call it a covariance.

- The one exception is Level 0, where we condition on the realized genotypes themselves. There $E[y_i \mid \text{genotypes}] = g_i \neq 0$, so the genuine object is the cross-moment $g_i g_j$, not a covariance (indeed $\text{Cov}[y_i, y_j \mid \text{genotypes}] = 0$, since only the independent environments remain).

Using (1) and [A2], every cross term involving e vanishes in expectation, so the phenotypic cross-moment reduces to the genetic one at every level:

$$C_{ij}(\mathcal{I}) = E[(g_i + e_i)(g_j + e_j) \mid \mathcal{I}] = E[g_i g_j \mid \mathcal{I}]. \quad (3)$$

The rest of the article evaluates (3) at four levels of resolution for $\mathcal{I}$.

1.3 Level 0 — the exact genetic cross-moment

At the finest resolution we condition on the realized genotypes of i and j at the causal variants. Then g_i and g_j are fixed, and by [A1],

$$C_{ij} = g_i g_j = \sum_k \sum_{k'} \beta_k \beta_{k'} x_{ik} x_{jk'}. \quad (4)$$

This is exact given only additivity. Notably it makes no reference to relatedness or IBD at all: the genetic covariance between two individuals is, fundamentally, an effect-size-weighted function of the genotypes they actually carry at causal sites. Everything below is a controlled approximation to (4) using progressively coarser — and more readily available — information.

1.4 Level 1 — dropping the effect weighting

Suppose we no longer know the individual genotypes, but do know the pair’s realized *sharing pattern* $\mathcal{S}$ (which alleles are IBD). We average (4) over the effect sizes.

[A3] Random, exchangeable causal effects.

Causal effects are drawn independently and identically across variants, with

$$E[\beta_k] = 0, \quad \text{Var}[\beta_k] = \sigma_\beta^2, \quad \beta \perp \{x\},$$

and their distribution does not depend on a variant’s genomic position or sharing.

Commonly bundled into: the infinitesimal / random-effects model; effect sizes independent of allele frequency and of local LD.

Taking the expectation of (4) over the effects, and using $\beta \perp \{x\}$ to factor:

$$E[g_i g_j \mid \mathcal{S}] = \sum_k \sum_{k'} E[\beta_k \beta_{k'}] E[x_{ik} x_{jk'} \mid \mathcal{S}].$$

Because the effects are i.i.d. with mean zero, $E[\beta_k \beta_{k'}] = 0$ whenever $k \neq k'$ and σ_β^2 when $k = k'$; the entire off-diagonal ($k \neq k'$) collapses regardless of any correlation between the genotypes. In other words, [A3] does the work that a separate “linkage equilibrium” assumption would otherwise do — we do not need one. What remains is

$$E[g_i g_j \mid \mathcal{S}] = \sigma_\beta^2 \sum_k E[x_{ik} x_{jk} \mid \mathcal{S}].$$

The variance bookkeeping fixes σ_β^2 : since $\text{Var}[x_{ik}] = 1$, we have $V_A = \sum_k E[\beta_k^2] \text{Var}[x_{ik}] = M\sigma_\beta^2$, hence $\sigma_\beta^2 = V_A/M$. Therefore

$$C_{ij}(\mathcal{S}) = V_A \cdot \frac{1}{M} \sum_k E[x_{ik} x_{jk} \mid \mathcal{S}]. \quad (5)$$

The per-variant effect weighting of (4) has been replaced by a flat average over causal variants of the per-variant genotype cross-moment.

1.5 Level 2 — realized relatedness

We now evaluate the per-variant cross-moment $E[x_{ik} x_{jk} \mid \mathcal{S}]$ in terms of IBD. It is cleanest to work at the level of individual alleles. At variant k (with population allele frequency p_k), let individual i carry the two allele indicators $a_{ik}^{(m)}, a_{ik}^{(p)} \in \{0, 1\}$ (maternal and paternal), each Bernoulli(p_k), so that the allele dosage is $G_{ik} = a_{ik}^{(m)} + a_{ik}^{(p)}$ and

$$x_{ik} = \frac{G_{ik} - 2p_k}{\sqrt{2p_k(1 - p_k)}}.$$

Two alleles are IBD if they descend from — and are copies of — the same ancestral allele. Let $\text{IBD}_{ij}(k) \in \{0, 1, 2\}$ denote the number of alleles individual i shares IBD with j at variant k .

[A5] No mutation (IBD $\Rightarrow$ IBS).

Alleles that are identical-by-descent are also identical-by-state: descent is transmitted without mutation.

Commonly bundled into: the infinite-sites / no-recurrent-mutation assumption.

[A6] Independence of non-IBD alleles.

Any two alleles that are *not* IBD are independent draws from the population allele distribution.

Commonly bundled into: within an individual, no inbreeding (Hardy–Weinberg); between individuals, random mating, no population structure, and no assortative mating. We use only the weak consequence stated here.

Under [A6] the two alleles within an individual are independent, so $\text{Var}[G_{ik}] = 2p_k(1-p_k)$ and the standardization indeed gives $\text{Var}[x_{ik}] = 1$. For the pair, conditioning on descent leaves the marginal allele distributions unchanged, so $\text{E}[x_{ik}] = 0$ and $\text{E}[x_{ik}x_{jk} \mid \mathcal{S}] = \text{Cov}[G_{ik}, G_{jk} \mid \mathcal{S}] / (2p_k(1-p_k))$. Expanding the genotype covariance over the four ordered allele pairs,

$$\text{Cov}[G_{ik}, G_{jk} \mid \mathcal{S}] = \sum_{u \in \{m,p\}} \sum_{v \in \{m,p\}} \text{Cov}[a_{ik}^{(u)}, a_{jk}^{(v)}],$$

each term is $\text{Var}[a] = p_k(1-p_k)$ if that allele pair is IBD (by [A5] it is the same copy, correlation 1) and 0 if it is not (by [A6], independent). If the pair shares $\text{IBD}_{ij}(k) = s$ alleles IBD, then exactly s of the four terms are nonzero, giving $\text{Cov}[G_{ik}, G_{jk} \mid s] = s p_k(1-p_k)$ and hence the key per-variant identity

$$\text{E}[x_{ik}x_{jk} \mid \text{IBD}_{ij}(k) = s] = \frac{s}{2} \quad (s = 0, 1, 2). \quad (6)$$

That is, a variant contributes correlation 0, $\frac{1}{2}$, or 1 according to whether the pair shares 0, 1, or 2 alleles IBD there.

To turn the causal-variant average in (5) into a genome-wide quantity we need one more assumption.

[A4] Homogeneous genome sharing.

The expected IBD sharing is the same at the causal variants as it is genome-wide; IBD is not enriched at particular loci.

Commonly bundled into: causal variants are representative of the genome with respect to relatedness.

Substituting (6) into (5) and applying [A4],

$$\frac{1}{M} \sum_k \text{E}[x_{ik}x_{jk} \mid \mathcal{S}] = \frac{1}{M} \sum_k \frac{\text{IBD}_{ij}(k)}{2} \xrightarrow{[A4]} \pi_{ij},$$

where π_{ij} is the *realized relatedness* — the proportion of the pair’s genome that is IBD. We arrive at the central Level-2 result:

$$\boxed{\text{E}[y_i y_j \mid \pi_{ij}] = \pi_{ij} V_A}. \quad (7)$$

The phenotypic covariance between two relatives is their realized fraction of shared genome times the additive variance. Because different pairs of the same nominal relationship share different realized fractions (full siblings scatter around $\frac{1}{2}$, and so on), (7) is expressed through the pair’s *actual* π_{ij} .

1.6 Level 3 — pedigree degree

Often we do not know π_{ij} but only the pair’s pedigree relationship (parent–offspring, full siblings, first cousins, ...).

[A7] Mendelian pedigree.

The relationship degree determines the distribution of IBD sharing through Mendelian segregation, so that $E[\pi_{ij} \mid \text{degree}]$ is the pedigree (expected) relatedness.

Commonly bundled into: a known pedigree with Mendelian inheritance.

Applying the tower property to (7) and averaging over the Mendelian distribution of π_{ij} given the degree d ,

$$E[y_i y_j \mid d] = E[E[y_i y_j \mid \pi_{ij}] \mid d] = V_A E[\pi_{ij} \mid d] = r_d V_A, \quad (8)$$

where $r_d = E[\pi_{ij} \mid d] = 2\phi_{ij}$ is the expected relatedness (twice the kinship coefficient). This recovers the classical covariances between relatives:

Relationship	Expected relatedness r_d
Monozygotic twins	1
Parent–offspring	1/2
Full siblings	1/2
Half sibs / grandparent–grandchild / avuncular	1/4
First cousins	1/8
“Unrelated”	0

The step from (7) to (8) is exactly the realized-versus-expected distinction: Level 2 uses a pair’s true realized relatedness, whereas Level 3 replaces it with its pedigree expectation, and is therefore only as good as the pedigree and the (often large) sampling variance of π_{ij} around r_d allow — a gap that widens for distant relationships.

1.7 Summary of the tower

Each level conditions on coarser information than the one above and is its conditional expectation; the rightmost column lists the assumption(s) newly required to descend one step.

Level	Conditioned on	$E[y_i y_j \mid \cdot]$	New assumptions
0	causal genotypes	$g_i g_j$	[A1] (given [A2])
1	realized sharing $\mathcal{S}$	$V_A \frac{1}{M} \sum_k E[x_{ik} x_{jk} \mid \mathcal{S}]$	[A3]
2	realized relatedness π_{ij}	$\pi_{ij} V_A$	[A4], [A5], [A6]
3	pedigree degree d	$r_d V_A$	[A7]

Next architectures restart from Level 0 and re-introduce a subset of [A1]–[A7], relaxing others: e.g. a shared environment breaks [A2]; dominance/epistasis breaks [A1]; assortative mating and population structure break [A6] (correlating non-IBD alleles) and drive a wedge between realized and pedigree relatedness.

References